

# Cryptography Notes

Alessandro Vulcu

4 novembre 2025

# Indice

|           |                                                                              |           |
|-----------|------------------------------------------------------------------------------|-----------|
| <b>1</b>  | <b>Divisibility</b>                                                          | <b>4</b>  |
| 1.1       | Introduction to Divisibility . . . . .                                       | 4         |
| 1.2       | Fundamental Properties and Proof . . . . .                                   | 4         |
| 1.3       | Even and Odd Integers . . . . .                                              | 5         |
| <b>2</b>  | <b>The Euclidean Division Algorithm</b>                                      | <b>5</b>  |
| 2.1       | The Division Theorem . . . . .                                               | 5         |
| 2.2       | Practical Computation of Quotient and Remainder . . . . .                    | 5         |
| <b>3</b>  | <b>Binary Representation of Natural Numbers</b>                              | <b>6</b>  |
| 3.1       | Definition and Process . . . . .                                             | 6         |
| 3.2       | Theorem on Binary Representation and Proof . . . . .                         | 7         |
| 3.3       | Properties of Binary Representation . . . . .                                | 8         |
| <b>4</b>  | <b>The Greatest Common Divisor (GCD)</b>                                     | <b>9</b>  |
| 4.1       | Definition of the GCD . . . . .                                              | 9         |
| 4.2       | The Euclidean Algorithm . . . . .                                            | 10        |
| <b>5</b>  | <b>The Extended Euclidean Algorithm and Diophantine Equations</b>            | <b>11</b> |
| 5.1       | Theorem and Application . . . . .                                            | 11        |
| 5.2       | Solving Linear Diophantine Equations . . . . .                               | 12        |
| <b>6</b>  | <b>Coprime Integers</b>                                                      | <b>14</b> |
| 6.1       | Definition and Characterization . . . . .                                    | 14        |
| 6.2       | The Prime Divisor Property . . . . .                                         | 14        |
| <b>7</b>  | <b>Conclusion: The Fundamental Theorem of Arithmetic</b>                     | <b>15</b> |
| <b>8</b>  | <b>Modular Arithmetic: Divisibility, Congruence, and Equivalence Classes</b> | <b>16</b> |
| <b>9</b>  | <b>The Principle of Divisibility and Its Relation to Congruence</b>          | <b>16</b> |
| 9.1       | Formal Definition of Divisibility . . . . .                                  | 16        |
| 9.2       | Illustrative Examples . . . . .                                              | 16        |
| <b>10</b> | <b>The Euclidean Division Algorithm</b>                                      | <b>17</b> |
| 10.1      | The Division Theorem . . . . .                                               | 17        |
| 10.2      | Components of the Equation . . . . .                                         | 17        |
| <b>11</b> | <b>The Theory of Congruence</b>                                              | <b>17</b> |
| 11.1      | Formal Definition of Congruence . . . . .                                    | 17        |
| 11.2      | Illustrative Examples . . . . .                                              | 18        |
| <b>12</b> | <b>Congruence as an Equivalence Relation</b>                                 | <b>18</b> |
| 12.1      | Definition of an Equivalence Relation . . . . .                              | 18        |
| <b>13</b> | <b>Equivalence Classes and Partitions</b>                                    | <b>19</b> |
| 13.1      | Illustrative Examples . . . . .                                              | 19        |

|                                                                                                  |           |
|--------------------------------------------------------------------------------------------------|-----------|
| <b>14 Conclusion</b>                                                                             | <b>20</b> |
| <b>15 A Comprehensive Treatise on Modular Arithmetic and Linear Congruences</b>                  | <b>21</b> |
| <b>16 The Foundation: Integers Modulo <math>m</math> (<math>\mathbb{Z}/m\mathbb{Z}</math>)</b>   | <b>21</b> |
| 16.1 The Congruence Relation . . . . .                                                           | 21        |
| 16.2 Constructing the Quotient Set $\mathbb{Z}/m\mathbb{Z}$ . . . . .                            | 21        |
| 16.3 Operations in $\mathbb{Z}/m\mathbb{Z}$ . . . . .                                            | 22        |
| <b>17 The Algebraic Landscape of <math>\mathbb{Z}/m\mathbb{Z}</math></b>                         | <b>22</b> |
| 17.1 The Additive Group $(\mathbb{Z}/m\mathbb{Z}, +)$ . . . . .                                  | 22        |
| 17.2 The Commutative Ring $(\mathbb{Z}/m\mathbb{Z}, +, \cdot)$ . . . . .                         | 23        |
| <b>18 Multiplicative Inverses and the Group of Units <math>(\mathbb{Z}/m\mathbb{Z})^*</math></b> | <b>23</b> |
| 18.1 Defining and Characterizing Invertibility . . . . .                                         | 23        |
| 18.2 Finding the Inverse . . . . .                                                               | 24        |
| 18.3 The Group of Units $(\mathbb{Z}/m\mathbb{Z})^*$ . . . . .                                   | 24        |
| <b>19 The Special Case of Prime Moduli: The Field <math>\mathbb{F}_p</math></b>                  | <b>24</b> |
| 19.1 Definition and Properties of a Field . . . . .                                              | 25        |
| 19.2 A Key Property of Fields . . . . .                                                          | 25        |
| <b>20 Application I: Solving Linear Congruences</b>                                              | <b>26</b> |
| 20.1 The General Condition for Solutions . . . . .                                               | 26        |
| 20.2 Case 1: Unique Solution ( $\gcd(a, m) = 1$ ) . . . . .                                      | 26        |
| 20.3 Case 2: Multiple or No Solutions ( $\gcd(a, m) = d > 1$ ) . . . . .                         | 26        |
| <b>21 Application II: Shamir's Secret Sharing Scheme</b>                                         | <b>26</b> |
| 21.1 The Premise and Analogy . . . . .                                                           | 27        |
| 21.2 A Worked Example . . . . .                                                                  | 27        |
| <b>22 An Exposition on the Chinese Remainder Theorem</b>                                         | <b>28</b> |
| <b>23 Introduction: An Ancient Problem with Modern Relevance</b>                                 | <b>28</b> |
| <b>24 The System of Congruences: Formalizing the Problem</b>                                     | <b>28</b> |
| <b>25 The Chinese Remainder Theorem: A Formal Statement</b>                                      | <b>28</b> |
| <b>26 Solution Methodologies</b>                                                                 | <b>29</b> |
| 26.1 Method 1: Iterative Substitution ("By Hands") . . . . .                                     | 29        |
| 26.2 Method 2: The Constructive Algorithm . . . . .                                              | 30        |
| <b>27 Formal Proof of the Chinese Remainder Theorem</b>                                          | <b>31</b> |
| 27.1 Proof of Existence . . . . .                                                                | 31        |
| 27.2 Proof of Uniqueness . . . . .                                                               | 31        |
| <b>28 An Important Extension: The Case of Non-Co-prime Moduli</b>                                | <b>31</b> |
| <b>29 Conclusion</b>                                                                             | <b>32</b> |

# 1 Divisibility

Technical Report on Divisibility, Euclidean Division, and Number Representation

## 1.1 Introduction to Divisibility

Divisibility is a fundamental pillar of number theory, providing the essential language for describing the relationships between integers. This concept serves as the bedrock upon which more advanced topics are built, including the computation of greatest common divisors via the Euclidean algorithm, the unique factorization of integers into primes, and the representation of numbers in different bases. This section formally defines this core relationship and explores its most immediate and essential properties.

---

### Definition of Divisibility

The relationship of divisibility is formally defined as follows: Let  $a, b$  be integers, with  $b \neq 0$ . We say that  $b$  divides  $a$  (written as  $b \mid a$ ) if there exists an integer  $c$  such that  $a = bc$ . For instance, we can state that  $2 \mid 4$  because  $4 = 2 \times 2$ . Conversely, 2 does not divide 5, as there is no integer  $c$  for which  $5 = 2c$ . It is critical to note that the definition explicitly prohibits division by zero; the divisor  $b$  must be a non-zero integer. This leads to a foundational question: “Does 1 divide any integer?” Based on the definition, the answer is yes. For any integer  $a$ , we can write  $a = 1 \times a$ , satisfying the condition with  $c = a$ . These elementary definitions give rise to several essential properties that govern the arithmetic of integers.

## 1.2 Fundamental Properties and Proof

The formal definition of divisibility leads directly to a set of properties that are instrumental in number theory.

**Proposition 1.1.** *Let  $a, b$ , and  $c$  be integers. The following properties hold:*

1. *If  $a \mid 1$ , then  $a = \pm 1$ .*
  2. *If  $a$  and  $b$  are non-zero integers such that  $a \mid b$  and  $b \mid a$ , then  $a = \pm b$ .*
  3. *If  $a \mid b$  and  $b \mid c$ , then  $a \mid c$  (Transitivity).*
- 

*Dimostrazione.* 1. If  $a \mid 1$ : By definition, there is an integer  $c$  such that  $1 = ac$ . Since  $a$  and  $c$  are integers, the only possible integer solutions are  $(a = 1, c = 1)$  or  $(a = -1, c = -1)$ . Therefore,  $a = \pm 1$ .

2. If  $a \mid b$  and  $b \mid a$ : By definition, there exist integers  $c_1$  and  $c_2$  such that  $b = ac_1$  and  $a = bc_2$ . Substituting the first equation into the second gives  $a = (ac_1)c_2 = a(c_1c_2)$ . Since  $a \neq 0$ , we can divide by  $a$  to get  $1 = c_1c_2$ . As  $c_1$  and  $c_2$  are integers, the only solutions are  $(c_1 = 1, c_2 = 1)$  or  $(c_1 = -1, c_2 = -1)$ .

- If  $c_1 = 1$ , then  $b = a \times 1 = a$ .

- If  $c_1 = -1$ , then  $b = a \times (-1) = -a$ .

Thus,  $a = \pm b$ .

3. If  $a \mid b$  and  $b \mid c$ : By definition, there exist integers  $c_1$  and  $c_2$  such that  $b = ac_1$  and  $c = bc_2$ . Substituting the first equation into the second gives  $c = (ac_1)c_2 = a(c_1c_2)$ . Since  $c_1c_2$  is an integer, this satisfies the definition of divisibility, so  $a \mid c$ . □

This demonstrates the transitive nature of the divisibility relation. A particularly common instance of divisibility is divisibility by 2, which gives rise to its own specific terminology.

### 1.3 Even and Odd Integers

An even number is an integer that is divisible by 2; an integer that is not divisible by 2 is an odd number. For example,  $-4$  is even because  $-4 = 2 \times (-2)$ , whereas  $13$  is odd. By convention, when we refer to the “divisors” of a natural number, we are typically referring to its positive divisors. For example, while it is true that  $-2 \mid 6$ , the set of divisors of  $6$  is conventionally listed as  $\{1, 2, 3, 6\}$ . When an integer does not divide another perfectly, the concept of a remainder becomes crucial. This idea is formalized by the Euclidean division algorithm.

## 2 The Euclidean Division Algorithm

### 2.1 The Division Theorem

Euclidean division is a cornerstone of number theory that generalizes the concept of divisibility. It provides a formal and precise description of the outcome of integer division, asserting that dividing any integer  $a$  by a positive integer  $b$  yields a unique integer quotient  $q$  and a unique integer remainder  $r$ .

**Proposition 2.1** (Euclidean Division). *Let  $a$  and  $b$  be natural numbers with  $b > 0$ . Then there exist unique integers  $q$  (the quotient) and  $r$  (the remainder) such that  $a = bq + r$ , with  $0 \leq r < b$ .*

A direct and important consequence of this theorem is that  $b$  divides  $a$  if and only if the remainder  $r = 0$ . For example, if  $a = 27$  and  $b = 7$ , we can write  $27 = 7 \times 3 + 6$ . Here, the quotient  $q$  is 3 and the remainder  $r$  is 6, which satisfies the condition  $0 \leq 6 < 7$ . While the theorem guarantees the existence and uniqueness of  $q$  and  $r$ , the next section details the practical method for their calculation.

### 2.2 Practical Computation of Quotient and Remainder

The quotient and remainder from Euclidean division can be computed directly using standard arithmetic operations. Starting with the identity  $a/b = q + r/b$ , we know from the theorem that  $0 \leq r < b$ , which implies  $0 \leq r/b < 1$ . This leads to the conclusion that the quotient  $q$  is the integer part of the fraction  $a/b$ . This is formally expressed using the floor function:

$$\text{Quotient: } q = \lfloor a/b \rfloor$$

Once the quotient is found, the remainder can be derived from the original equation  $a = bq + r$ :

$$\text{Remainder: } r = a - bq = a - b \times \lfloor a/b \rfloor$$

To illustrate, let  $a = 15476$  and  $b = 137$ .

- Calculate the ratio:  $15476/137 \approx 112.96 \dots$
- The quotient is the floor of this value:  $q = \lfloor 112.96 \dots \rfloor = 112$ .
- The remainder is then:  $r = 15476 - 137 \times 112 = 15476 - 15344 = 132$ .

The repeated application of this division process is the key to representing numbers in different bases, most notably in the binary system.

## 3 Binary Representation of Natural Numbers

### 3.1 Definition and Process

Binary representation is a system for expressing any natural number as a sum of powers of 2, using only the digits 0 and 1. This base-2 system is foundational to all modern digital computing. The algorithm to convert a number to its binary form is a direct application of the principles of Euclidean division.

#### Definition

Let  $m \in \mathbb{N}$ . We write  $m = (m_{B-1} \dots m_1 m_0)_2$  if  $m = m_0 + m_1 \times 2 + \dots + m_{B-1} 2^{B-1}$  and  $m_i \in \{0, 1\}$ . The values  $m_i$  are known as bits. The process for converting a number to binary involves repeatedly dividing the number by 2 and recording the remainders. The remainders, read in reverse order of their calculation, form the binary representation.

Tabella 1: Conversion of Decimal 112 to Binary

| Number ( $m$ ) | Quotient ( $q$ ) | Remainder ( $r$ ) = Bit |
|----------------|------------------|-------------------------|
| $112 \div 2$   | 56               | 0 ( $m_0$ )             |
| $56 \div 2$    | 28               | 0 ( $m_1$ )             |
| $28 \div 2$    | 14               | 0 ( $m_2$ )             |
| $14 \div 2$    | 7                | 0 ( $m_3$ )             |
| $7 \div 2$     | 3                | 1 ( $m_4$ )             |
| $3 \div 2$     | 1                | 1 ( $m_5$ )             |
| $1 \div 2$     | 0                | 1 ( $m_6$ )             |

Reading the remainders from bottom to top, we find that 112 in decimal is  $(1110000)_2$  in binary. This algorithmic process is guaranteed to terminate and provides a unique representation for any natural number, as established by the following theorem.

## 3.2 Theorem on Binary Representation and Proof

**Proposition 3.1.** *Let  $m \in \mathbb{N}$ , with  $m < 2^B$ . The repeated division-by-2 process:*

$$m = 2q_0 + m_0$$

$$q_0 = 2q_1 + m_1$$

$$q_1 = 2q_2 + m_2$$

...

*terminates with  $q_{B-1} = 0$ , and  $m = m_0 + 2m_1 + \dots + 2^{B-1}m_{B-1}$ .*

*Dimostrazione.* The proof consists of two parts: demonstrating the polynomial form and proving that the process terminates.

### Polynomial Form

We can show the relationship by algebraic substitution.

$$m = 2q_0 + m_0$$

Substitute  $q_0 = 2q_1 + m_1$ :

$$m = 2(2q_1 + m_1) + m_0 = 2^2q_1 + 2m_1 + m_0$$

Substitute  $q_1 = 2q_2 + m_2$ :

$$m = 2^2(2q_2 + m_2) + 2m_1 + m_0 = 2^3q_2 + 2^2m_2 + 2m_1 + m_0$$

Continuing this process until the final step where  $q_{B-2} = 2q_{B-1} + m_{B-1}$ , we arrive at the final form:  $m = m_0 + 2m_1 + 2^2m_2 + \dots + 2^{B-1}m_{B-1}$ .

### Termination

The process is guaranteed to terminate because the sequence of quotients is strictly decreasing and bounded.

- Since  $m < 2^B$ , the first division  $m = 2q_0 + m_0$  implies  $2q_0 \leq m < 2^B$ , so  $q_0 < 2^{B-1}$ .
- Similarly,  $q_0 < 2^{B-1}$  implies  $2q_1 \leq q_0 < 2^{B-1}$ , so  $q_1 < 2^{B-2}$ .

This pattern continues:  $q_i < 2^{B-1-i}$ . For the final quotient  $q_{B-1}$ , we have  $q_{B-1} < 2^{B-1-(B-1)} = 2^0 = 1$ . Since  $q_{B-1}$  is a non-negative integer, the only possibility is  $q_{B-1} = 0$ . The process terminates when the quotient becomes zero.  $\square$

This theorem also establishes a direct relationship between a number's magnitude and the number of bits required to represent it.

### 3.3 Properties of Binary Representation

**Remark.** If  $m$  has  $B$  bits in its binary representation, i.e.,  $m = (m_{B-1} \dots m_0)_2$ , then  $m < 2^B$ .

*Dimostrazione.* The largest possible value for a  $B$ -bit number occurs when all bits are 1. This value can be calculated by summing the geometric series:

$$m \leq 1 + 2 + 4 + \dots + 2^{B-1} = \frac{2^B - 1}{2 - 1} = 2^B - 1$$

Therefore,  $m < 2^B$ . □

---

This leads to a corollary that precisely characterizes the number of bits required for a given integer.

**Corollary 3.1.** A natural number  $m$  has a binary representation with  $B$  bits (where the most significant bit  $m_{B-1} = 1$ ) if and only if  $2^{B-1} \leq m < 2^B$ .

This relationship provides a practical formula for finding the required number of bits,  $B$ , for any number  $m$  using logarithms:

$$B = \lfloor \log_2 m \rfloor + 1$$

For example, to find the number of bits required to represent the number 368932:

- Calculate the base-2 logarithm:  $\log_2(368932) \approx 18.4 \dots$
- Apply the formula:  $B = \lfloor 18.4 \dots \rfloor + 1 = 18 + 1 = 19$ . It requires 19 bits.

The Euclidean division algorithm is not only useful for number representation but is also the most efficient method for finding the greatest common divisor of two integers.



## 4 The Greatest Common Divisor (GCD)

### 4.1 Definition of the GCD

While divisibility focuses on a single divisor-dividend pair, the concept of the Greatest Common Divisor (GCD) analyzes the shared properties of divisibility between two or more integers. It is a concept of paramount importance in number theory, with applications ranging from simplifying fractions to modern cryptography.

#### Definition

Let  $a$  and  $b$  be integers, not both zero. The set of their common divisors has a largest element  $d$ , called the greatest common divisor of  $a$  and  $b$ . We write  $d = \gcd(a, b)$ . To find  $\gcd(18, 12)$ , we can list the divisors of each number:

- Divisors of 12:  $\{1, 2, 3, 4, 6, 12\}$
- Divisors of 18:  $\{1, 2, 3, 6, 9, 18\}$

The set of common divisors is  $\{1, 2, 3, 6\}$ . The largest element in this set is 6, so  $\gcd(18, 12) = 6$ . While listing all divisors is feasible for small numbers, it is impractical for large integers. A far more powerful and efficient method is the Euclidean Algorithm.

## 4.2 The Euclidean Algorithm

The Euclidean Algorithm is a systematic and highly efficient procedure for computing the GCD of two positive integers by using repeated Euclidean division.

**Theorem 4.1** (Euclidean Algorithm). *Let  $a, b > 0$  with  $a \geq b$ . The following algorithm computes  $\gcd(a, b)$  in a finite number of steps:*

1. Let  $r_0 := a, r_1 := b$ .
2. Set  $i = 1$ .
3. Divide  $r_{i-1}$  by  $r_i$  to get a quotient  $q_i$  and remainder  $r_{i+1}$ :  $r_{i-1} = r_i q_i + r_{i+1}$ .
4. If  $r_{i+1} = 0$ , then  $r_i = \gcd(a, b)$ . The algorithm terminates.
5. Otherwise, increment  $i$  and return to Step 3.

Let's apply the algorithm to find  $\gcd(259, 119)$ :

$$\begin{aligned} 259 &= 119 \times 2 + 21 & (r_2 = 21) \\ 119 &= 21 \times 5 + 14 & (r_3 = 14) \\ 21 &= 14 \times 1 + 7 & (r_4 = 7) \\ 14 &= 7 \times 2 + 0 & (r_5 = 0) \end{aligned}$$

The last non-zero remainder is 7. Therefore,  $\gcd(259, 119) = 7$ .

---

*Proof of Correctness.* Let the sequence of remainders be  $r_0, r_1, \dots, r_n, r_{n+1}$  where  $d = r_n$  is the last non-zero remainder and  $r_{n+1} = 0$ . The proof must show two things: that  $d$  is a common divisor of  $a$  and  $b$ , and that it is the greatest common divisor.

### a. $d$ divides $a$ and $b$

We proceed by working backward from the end of the algorithm.

- The final division step is  $r_{n-1} = r_n q_n + r_{n+1}$ . Since  $r_{n+1} = 0$  and  $d = r_n$ , this equation becomes  $r_{n-1} = d q_n$ . This demonstrates that  $d \mid r_{n-1}$ .
- The preceding step is  $r_{n-2} = r_{n-1} q_{n-1} + r_n$ . We know  $d \mid r_n$  (trivially) and we have just shown that  $d \mid r_{n-1}$ . Therefore,  $d$  must divide their linear combination, so  $d \mid r_{n-2}$ .

Continuing this process of backward induction, we establish that  $d$  divides every preceding remainder, until we conclude that  $d$  divides  $r_1$  (which is  $b$ ) and  $r_0$  (which is  $a$ ). Thus,  $d$  is a common divisor.

## b. $d$ is the greatest common divisor

Now let  $c$  be any common divisor of  $a$  and  $b$ . We work forward down the chain of remainders.

- Since  $c \mid a$  ( $r_0$ ) and  $c \mid b$  ( $r_1$ ), it must divide the first remainder  $r_2 = r_0 - r_1q_1$ .
- Since  $c \mid r_1$  and  $c \mid r_2$ , it must divide their remainder  $r_3 = r_1 - r_2q_2$ .

Continuing this process, we eventually find that  $c$  must divide the last non-zero remainder,  $r_n$ , which is  $d$ . Since  $c$  divides  $d$ , it must be that  $c \leq d$ . This holds for any common divisor  $c$ , proving that  $d$  is the greatest common divisor.  $\square$

---

This powerful algorithm can be extended not just to find the GCD, but also to express it as a specific integer combination of the original numbers.

## 5 The Extended Euclidean Algorithm and Diophantine Equations

### 5.1 Theorem and Application

The Extended Euclidean Algorithm is a crucial enhancement of the standard algorithm. It demonstrates that the GCD of two integers can always be expressed as a linear combination of those integers. This property is fundamental to number theory and is the key to solving linear Diophantine equations.

**Proposition 5.1** (Extended Euclidean Algorithm). *Let  $a$  and  $b$  be natural numbers, not both zero. If  $d = \gcd(a, b)$ , then there exist integers  $u, v$  such that  $d = au + bv$ .*

To find these coefficients  $u$  and  $v$ , we work backward through the steps of the standard Euclidean algorithm, substituting for the remainders at each stage. Using our previous result  $\gcd(259, 119) = 7$  (let  $a = 259, b = 119$ ):

From  $21 = 14 \times 1 + 7$ , we isolate the remainder:  $7 = 21 - 14$

From  $119 = 21 \times 5 + 14$ , we have  $14 = 119 - 21 \times 5$ . Substitute this:

$$7 = 21 - (119 - 21 \times 5) = 21 - 119 + 5 \times 21 = 6 \times 21 - 119$$

From  $259 = 119 \times 2 + 21$ , we have  $21 = 259 - 119 \times 2$ . Substitute this final expression:

$$7 = 6 \times (259 - 119 \times 2) - 119$$

$$7 = 6 \times 259 - 12 \times 119 - 119$$

$$7 = 6 \times 259 - 13 \times 119$$

Thus, we have found  $7 = 6a - 13b$ , where  $u = 6$  and  $v = -13$ . It is important to note that these coefficients  $u$  and  $v$  are not unique. Let  $(u, v)$  be one such pair of coefficients. Now, for any integer  $k$ , define  $u' = u + k(b/d)$  and  $v' = v - k(a/d)$ . (The original text uses  $u' = u + kb$  and  $v' = v - ka$ , which is correct only if  $d = 1$ , but the more general form is needed here). Let's

use the general form  $u' = u + k \frac{b}{\gcd(a,b)}$  and  $v' = v - k \frac{a}{\gcd(a,b)}$ . Substituting these into the linear combination yields:

$$\begin{aligned}
 au' + bv' &= a \left( u + k \frac{b}{d} \right) + b \left( v - k \frac{a}{d} \right) \\
 &= au + ak \frac{b}{d} + bv - bk \frac{a}{d} \\
 &= au + bv + k \left( \frac{ab}{d} - \frac{ab}{d} \right) \\
 &= au + bv = d
 \end{aligned}$$

This demonstrates that there are infinitely many pairs of coefficients. This theorem is the key to analyzing and solving a general class of equations known as linear Diophantine equations.

## 5.2 Solving Linear Diophantine Equations

A linear Diophantine equation is an equation of the form  $au + bv = c$ , where  $a, b, c$  are known integers and  $u, v$  are the integer variables to be solved for.

**Proposition 5.2** (Solvability Condition). *The equation  $au + bv = c$  has integer solutions  $u, v$  if and only if  $\gcd(a, b) \mid c$ .*

---

*Dimostrazione.*  $(\Rightarrow)$  Assume integer solutions  $u, v$  exist. Let  $d = \gcd(a, b)$ . By definition,  $d \mid a$  and  $d \mid b$ . Therefore,  $d$  must divide any linear combination of  $a$  and  $b$ , so  $d \mid (au + bv)$ . Since  $au + bv = c$ , it follows that  $d \mid c$ .

$(\Leftarrow)$  Assume  $\gcd(a, b) \mid c$ . Let  $d = \gcd(a, b)$ . Then  $c = kd$  for some integer  $k$ . From the Extended Euclidean Algorithm, we know there exist integers  $u_0, v_0$  such that  $au_0 + bv_0 = d$ . Multiplying by  $k$ , we get  $a(ku_0) + b(kv_0) = kd = c$ . Thus, a solution exists with  $u = ku_0$  and  $v = kv_0$ . □

### General Solution Methodology

To find all integer solutions to  $au + bv = c$ :

1. Calculate  $d = \gcd(a, b)$  using the Euclidean Algorithm. If  $d$  does not divide  $c$ , there are no integer solutions.
2. If  $d \mid c$ , divide the entire equation by  $d$  to get a simplified equation  $a'u + b'v = c'$ , where  $a' = a/d$ ,  $b' = b/d$ , and  $c' = c/d$ . Note that  $\gcd(a', b') = 1$ .
3. Use the Extended Euclidean Algorithm to find a particular solution  $(u_*, v_*)$  for the equation  $a'u + b'v = 1$ .
4. A particular solution  $(u_p, v_p)$  for the simplified equation from step 2 is then given by  $(u_*c', v_*c')$ . This provides a particular solution to the original equation as well.
5. The complete set of solutions is given by the general formula described below, using the particular solution  $(u_p, v_p)$  and the coprime values  $a'$  and  $b'$ .

**Proposition 5.3** (Form of All Solutions). *If  $(u_p, v_p)$  is a particular solution to  $au + bv = c$  and  $d = \gcd(a, b)$ , let  $a' = a/d$  and  $b' = b/d$ . Then all integer solutions are given by:*

$$u = u_p + kb' \quad \text{and} \quad v = v_p - ka'$$

*for any integer  $k$ .*

*Dimostrazione.* We have already shown that any pair  $(u, v)$  of this form is a solution (using  $a' = b/d$  and  $b' = a/d$  in the substitution check in section 5.1). Now, assume  $(u', v')$  is another solution, so  $au' + bv' = c$ . We also have  $au_p + bv_p = c$ . Subtracting the two equations gives  $a(u' - u_p) + b(v' - v_p) = 0$ , or  $a(u' - u_p) = b(v_p - v')$ . Dividing by  $d$ :  $a'(u' - u_p) = b'(v_p - v')$ . Since  $\gcd(a', b') = 1$ , and  $a'$  divides the right-hand side  $b'(v_p - v')$ , it must be that  $a' \mid (v_p - v')$ . Therefore,  $v_p - v' = ka'$  for some integer  $k$ , which means  $v' = v_p - ka'$ . Substituting this back gives  $a'(u' - u_p) = b'(ka')$ , which simplifies to  $u' - u_p = kb'$ , or  $u' = u_p + kb'$ . This shows that any other solution must be of the specified form.  $\square$

The special case where  $\gcd(a, b) = 1$  is of particular importance in number theory and has its own terminology.

## 6 Coprime Integers

### 6.1 Definition and Characterization

Coprime (or relatively prime) integers are those that share no common divisors other than 1 and  $-1$ . This relationship is central to many theorems in number theory, including those related to prime numbers and factorization.

#### Definition

Two integers  $a, b$  are relatively prime (or coprime) if  $\gcd(a, b) = 1$ . For example, 32 and 15 are relatively prime. The divisors of 32 are  $\{1, 2, 4, 8, 16, 32\}$  and the divisors of 15 are  $\{1, 3, 5, 15\}$ . Their only common positive divisor is 1.

**Proposition 6.1** (Characterisation of Coprime Numbers). *Integers  $a$  and  $b$  are relatively prime if and only if there exist integers  $u, v$  such that  $au + bv = 1$ .*

---

*Dimostrazione.* ( $\Rightarrow$ ) This is a direct consequence of the Extended Euclidean Algorithm. If  $\gcd(a, b) = 1$ , then the algorithm guarantees the existence of integers  $u, v$  such that  $au + bv = 1$ .

( $\Leftarrow$ ) Conversely, assume there are integers  $u, v$  such that  $au + bv = 1$ . Let  $d = \gcd(a, b)$ . Since  $d \mid a$  and  $d \mid b$ , it must divide their linear combination  $au + bv$ . Therefore,  $d \mid 1$ . The only positive integer that divides 1 is 1, so  $d = 1$ . □

---

This characterization leads directly to an essential property regarding prime numbers.

### 6.2 The Prime Divisor Property

**Corollary 6.1.** *Let  $p$  be a prime number and  $m$  be an integer. If  $p$  does not divide  $m$ , then  $p$  and  $m$  are coprime.*

*Dimostrazione.* The divisors of a prime number  $p$  are only  $\{1, p\}$ . If  $p$  does not divide  $m$ , then  $p$  is not a common divisor. Therefore, the only common divisor of  $p$  and  $m$  is 1, which means  $\gcd(p, m) = 1$ . □

---

This corollary is the key to proving the following critical proposition, which is commonly known as Euclid's Lemma.

**Proposition 6.2** (Prime Divisor Property). *Let  $p$  be a prime and  $a, b$  be integers. If  $p \mid ab$ , then  $p \mid a$  or  $p \mid b$ .*

*Dimostrazione.* Assume  $p \mid ab$ . If  $p \mid a$ , the proposition is true. If  $p$  does not divide  $a$ , then by the corollary above,  $\gcd(p, a) = 1$ . By the characterization of coprime numbers, there exist integers  $u, v$  such that  $pu + av = 1$ . Multiplying this equation by  $b$  gives  $pub + abv = b$ .

- We know  $p$  divides  $pub$ .
- We are given that  $p$  divides  $ab$ , so it must also divide  $abv$ .

Since  $p$  divides both terms on the left side, it must divide their sum. Therefore,  $p \mid b$ . □

---

This property is unique to prime numbers. For a non-prime (composite) number, this does not hold. For example,  $4 \mid 2 \times 6$  (since  $4 \mid 12$ ), but 4 does not divide 2 and 4 does not divide 6. This unique property of primes is the cornerstone of the most important theorem in elementary number theory.

## 7 Conclusion: The Fundamental Theorem of Arithmetic

This report has traced a logical path from the elementary definition of divisibility to the profound and unique properties of prime numbers. We began by establishing the rules of integer division, which we generalized with the Euclidean Division theorem. This theorem, in turn, provided a practical algorithm not only for finding quotients and remainders—enabling number system conversions like binary representation—but also for efficiently computing the greatest common divisor. By extending the Euclidean algorithm, we established that the GCD can be expressed as a linear combination, a result that unlocked the solution to linear Diophantine equations and provided a powerful characterization for coprime integers. The culmination of this journey is the Prime Divisor Property, which states that if a prime divides a product, it must divide one of the factors. This property, which we proved does not hold for composite numbers, is the essential lemma required to prove the Fundamental Theorem of Arithmetic. This theorem asserts that every integer greater than 1 is either a prime number itself or can be represented as a unique product of prime numbers, cementing the role of primes as the fundamental building blocks of the integers.

## 8 Modular Arithmetic: Divisibility, Congruence, and Equivalence Classes

### Introduction

Modular arithmetic is a system of arithmetic for integers where numbers “wrap around” after reaching a certain value, known as the **modulus**. This concept is intuitively understood through the analogy of “clock arithmetic.” On a standard 12-hour clock, the hours cycle from 1 to 12. If it is 9 o’clock, adding 6 hours results in 3 o’clock, not 15 o’clock. Similarly, 3 hours before 2 o’clock is 11 o’clock. In this system, the number 12 effectively plays the role of zero. Modular arithmetic formalizes this intuitive idea, creating a powerful mathematical structure. This document provides a formal, academic exploration of its foundational principles, rigorously defining divisibility, the Euclidean division algorithm, and congruence. This progression will culminate in an analysis of equivalence classes and partitions, which form the structural backbone of this essential mathematical system.

---

## 9 The Principle of Divisibility and Its Relation to Congruence

The concept of divisibility is the bedrock upon which modular arithmetic is built. Understanding the conditions under which one integer can be perfectly divided by another is the first and most critical step toward comprehending the structure of congruences. It allows us to define a baseline—a “zero point”—within any given modulus.

### 9.1 Formal Definition of Divisibility

An integer  $m$  is said to divide an integer  $a$ , denoted  $m \mid a$ , if there exists an integer  $k$  such that  $a = km$ . A direct and fundamental consequence of this definition is its relationship to congruence. An integer  $a$  is divisible by  $m$  if and only if  $a$  is congruent to zero modulo  $m$ .

$$m \mid a \text{ is equivalent to the statement } a \equiv 0 \pmod{m}$$

This equivalence establishes that all multiples of the modulus  $m$  are considered equivalent to zero within that modular system.

### 9.2 Illustrative Examples

- **Clock Arithmetic:** In a mod12 system, the number 12 plays the role of zero. This is formally expressed as  $12 \equiv 0 \pmod{12}$ , which is true because the difference  $12 - 0 = 12$  is a multiple of 12.
- **General Multiples:** The concept extends to any modulus  $m$ . All integer multiples of  $m$  (e.g.,  $m, 2m, -m$ , etc.) are congruent to 0  $\pmod{m}$ .

The case of exact divisibility, where the remainder is zero, is a specific instance of a more general principle. The next logical step is to consider the outcome of division when the remainder is not zero, which is formally described by the Euclidean Division Algorithm.

---



## 10 The Euclidean Division Algorithm

The **Euclidean Division Algorithm** is the essential mechanism for determining the relationship between any two integers within a given modulus. Its true power lies in its guarantee that for any integer  $a$  and modulus  $m$ , there exists a unique remainder  $r$  in the set  $\{0, 1, \dots, m-1\}$ . This uniqueness is the linchpin that allows every integer to be mapped to a canonical representative, which is the foundational operation of modular arithmetic.

### 10.1 The Division Theorem

**Proposition 10.1.** *For any integer  $a$  (the dividend) and any positive integer  $m > 1$  (the modulus or divisor), there exist unique integers  $q$  (the quotient) and  $r$  (the remainder) such that:*

$$a = mq + r$$

where the remainder  $r$  is constrained by the condition:

$$0 \leq r < m$$

### 10.2 Components of the Equation

| Term | Description                               |
|------|-------------------------------------------|
| $a$  | The dividend (the integer being divided). |
| $m$  | The divisor (the modulus).                |
| $q$  | The quotient.                             |
| $r$  | The remainder.                            |

The primary implication of this algorithm is that for any integers  $a$  and  $m$  (with  $m > 1$ ), there exists a unique remainder  $r$  that is an element of the set  $\{0, 1, \dots, m-1\}$ . This unique remainder is fundamentally linked to the original integer  $a$  through the concept of congruence, which will be detailed next.

---

## 11 The Theory of Congruence

Congruence is a powerful generalization of equality that forms the operational basis of modular arithmetic. It provides a systematic way to classify all integers based on their remainders with respect to a given modulus  $m$ . Two numbers are considered congruent if they are “the same” within that modulus, even if their absolute values are different.

### 11.1 Formal Definition of Congruence

Two integers  $a$  and  $b$  are **congruent modulo  $m$**  if their difference,  $a - b$ , is divisible by  $m$ . This is expressed with the notation:

$$a \equiv b \pmod{m}$$

This definition is entirely equivalent to the statement that  $a$  and  $b$  have the same remainder upon division by  $m$ . This equivalence can be formally shown:

( $\Rightarrow$ ) Assume  $a \equiv b \pmod{m}$ . Then  $a - b = km$  for some integer  $k$ . By the Division Algorithm,  $a = mq_1 + r_1$  and  $b = mq_2 + r_2$ , where  $0 \leq r_1, r_2 < m$ . Subtracting gives  $a - b = m(q_1 - q_2) + (r_1 - r_2)$ . Substituting  $km$  yields  $r_1 - r_2 = m(k - q_1 + q_2)$ . Since  $-m < r_1 - r_2 < m$ , the only integer multiple of  $m$  in this range is 0. Thus,  $r_1 - r_2 = 0$ , implying  $r_1 = r_2$ .

( $\Leftarrow$ ) Assume  $a$  and  $b$  have the same remainder  $r$ . Then  $a = mq_1 + r$  and  $b = mq_2 + r$ . Subtracting gives  $a - b = m(q_1 - q_2)$ . By definition,  $m \mid (a - b)$ , and thus  $a \equiv b \pmod{m}$ .

**Theorem 11.1** (Existence of a Unique Remainder). *For any integer  $a$  and modulus  $m > 1$ , there exists a unique integer  $r$  in the set  $\{0, 1, \dots, m - 1\}$  such that  $a \equiv r \pmod{m}$ .*

*Dimostrazione.* Begin with the Euclidean Division Algorithm:  $a = mq + r$ , where  $0 \leq r < m$ . Rearrange:  $a - r = mq$ . By the formal definition of divisibility,  $m \mid (a - r)$ . By the formal definition of congruence, this directly implies  $a \equiv r \pmod{m}$ .  $\square$

## 11.2 Illustrative Examples

- **Clock Arithmetic** ( $\pmod{12}$ ):

- $15 \equiv 3 \pmod{12}$  is true because  $15 - 3 = 12$ , which is a multiple of 12.
- $-1 \equiv 11 \pmod{12}$  is true because  $-1 - 11 = -12$ , which is a multiple of 12.

- **Numerical Example 1:**  $17 \equiv 7 \pmod{10}$  is true because  $17 - 7 = 10$ , which is a multiple of 10.
- **Numerical Example 2:**  $17 \equiv 3 \pmod{7}$  is true because  $17 - 3 = 14$ , and  $14 = 2 \times 7$ .

The properties of congruence are not arbitrary; they conform to a precise mathematical structure known as an equivalence relation, which ensures that congruence behaves logically like equality.

---

## 12 Congruence as an Equivalence Relation

The formal classification of congruence as an **equivalence relation** is of paramount importance. This classification is critical because it formally proves that congruence behaves with the same logical consistency as equality (*reflexivity, symmetry, transitivity*), which is the prerequisite for partitioning a set into meaningful classes.

### 12.1 Definition of an Equivalence Relation

An equivalence relation, denoted  $\sim$ , on a set  $X$  is a relation that satisfies the following three properties:

1. **Reflexive:** For all  $a \in X$ ,  $a \sim a$ .
2. **Symmetric:** For all  $a, b \in X$ , if  $a \sim b$  then  $b \sim a$ .
3. **Transitive:** For all  $a, b, c \in X$ , if  $a \sim b$  and  $b \sim c$ , then  $a \sim c$ .

**Theorem 12.1** (Congruence is an Equivalence Relation). *The relation of congruence modulo  $m$  is an equivalence relation on the set of integers  $\mathbb{Z}$ .*

*Dimostrazione.* We must demonstrate that congruence modulo  $m$  satisfies all three properties.

- **Proof of Reflexivity** For any integer  $a$ ,  $a - a = 0$ . Since  $m \mid 0$ ,  $a \equiv a \pmod{m}$ . The relation is reflexive.
- **Proof of Symmetry** Assume  $a \equiv b \pmod{m}$ . Then  $m \mid (a - b)$ , so  $a - b = km$ . Multiplying by  $-1$  gives  $b - a = (-k)m$ . Since  $m \mid (b - a)$ ,  $b \equiv a \pmod{m}$ . The relation is symmetric.
- **Proof of Transitivity** Assume  $a \equiv b \pmod{m}$  and  $b \equiv c \pmod{m}$ . Then  $a - b = k_1m$  and  $b - c = k_2m$ . Adding the equations yields:

$$(a - b) + (b - c) = k_1m + k_2m$$

$$a - c = (k_1 + k_2)m$$

Since  $k_1 + k_2$  is an integer,  $m \mid (a - c)$ , which implies  $a \equiv c \pmod{m}$ . The relation is transitive.

Having satisfied all three properties, the proof that congruence modulo  $m$  is an equivalence relation is complete.  $\square$

## 13 Equivalence Classes and Partitions

The ultimate consequence of an equivalence relation is the partitioning of a set into disjoint subsets, known as **equivalence classes**. These classes are fundamental because they group all elements that are considered “the same” with respect to the relation. In modular arithmetic, these classes group all integers that share the same remainder when divided by the modulus  $m$ .

### Formal Definitions

- **The equivalence class** of an element  $a$  in a set  $X$ , denoted  $[a]$ , is the set of all elements  $b$  in  $X$  that are related to  $a$ :

$$[a] := \{b \in X : b \sim a\}$$

- **The quotient set**, denoted  $X / \sim$ , is the set of all equivalence classes.

### 13.1 Illustrative Examples

- **Same Parity (Integers Modulo 2):** This equivalence relation on  $\mathbb{Z}$  produces exactly two equivalence classes:

$$- [0] = \{\dots, -4, -2, 0, 2, 4, \dots\} \text{ (The set of all even numbers)}$$

$$- [1] = \{\dots, -3, -1, 1, 3, 5, \dots\} \text{ (The set of all odd numbers)}$$

- **Human Beings with the Same Parents:** The class  $[Ed]$  is the set of all human beings who have both of Ed’s biological parents (Ed and all of their full siblings).

**Proposition 13.1** (Equivalence Classes Form a Partition). *Let  $\sim$  be an equivalence relation on a set  $X$ . The set of equivalence classes,  $X / \sim$ , forms a partition of  $X$ .*

For a collection of subsets to be a partition, two conditions must be met:

1. The union of all equivalence classes is equal to the original set  $X$ .
2. Any two distinct equivalence classes are disjoint (their intersection is the empty set).

## 14 Conclusion

The principles of modular arithmetic unfold in a sequence of remarkable logical dependency. We began with the primitive notion of integer divisibility, which served as the cornerstone for the Euclidean Division Algorithm. This algorithm's guarantee of a unique remainder, in turn, allowed for the rigorous definition of congruence—a powerful generalization of equality. Finally, by proving that congruence satisfies the properties of an equivalence relation, we unlocked its ultimate structural consequence: the partitioning of the infinite set of integers into a finite, manageable collection of residue classes. These foundational principles are not only elegant but are essential for further study in abstract algebra, number theory, and computer science.

## 15 A Comprehensive Treatise on Modular Arithmetic and Linear Congruences

Modular arithmetic is a system of arithmetic for integers where numbers “wrap around” upon reaching a certain value—the modulus. This elegant branch of number theory simplifies problems involving large or infinite sets of integers by transforming them into a finite, manageable system. Its strategic importance is profound, providing the foundational mathematics for fields as diverse as cryptography, computer science, and abstract algebra, where it allows for the analysis of complex structures through their finite analogues.

## 16 The Foundation: Integers Modulo $m$ ( $\mathbb{Z}/m\mathbb{Z}$ )

The cornerstone of modular arithmetic is the concept of congruence, which formalizes the notion of numbers being “the same” with respect to a modulus.

### 16.1 The Congruence Relation

**Definition 16.1.** *Let  $m$  be an integer greater than 1. Two integers,  $a$  and  $b$ , are said to be **congruent modulo  $m$**  if their difference  $(a - b)$  is divisible by  $m$ . This relationship is denoted as:*

$$a \equiv b \pmod{m} \quad \text{which is formally defined as} \quad m \mid (a - b)$$

This congruence relation is an equivalence relation on the set of integers,  $\mathbb{Z}$ , meaning it partitions the entire set of integers into disjoint subsets, known as equivalence classes. An equivalence relation must satisfy three fundamental properties:

1. **Reflexivity:**  $a \equiv a \pmod{m}$  for any integer  $a$ .
2. **Symmetry:** If  $a \equiv b \pmod{m}$ , then  $b \equiv a \pmod{m}$ .
3. **Transitivity:** If  $a \equiv b \pmod{m}$  and  $b \equiv c \pmod{m}$ , then  $a \equiv c \pmod{m}$ .

### 16.2 Constructing the Quotient Set $\mathbb{Z}/m\mathbb{Z}$

The congruence relation partitions the set of integers  $\mathbb{Z}$  into distinct equivalence classes.

- An **equivalence class**  $[a]$  is the set of all integers congruent to  $a$  modulo  $m$ :

$$[a] = \{b \in \mathbb{Z} \mid b \equiv a \pmod{m}\}$$

- The **quotient set**, denoted  $\mathbb{Z}/m\mathbb{Z}$ , is the set of all such equivalence classes modulo  $m$ .

Critically, the set  $\mathbb{Z}/m\mathbb{Z}$  contains exactly  $m$  distinct elements. Any integer is congruent to its remainder upon division by  $m$ , which must be a value from 0 to  $m - 1$ . Consequently, the  $m$  distinct classes are  $[0], [1], [2], \dots, [m - 1]$ . For notational simplicity, it is common practice to represent these classes by the integers themselves:  $\mathbb{Z}/m\mathbb{Z} = \{0, 1, \dots, m - 1\}$ . It is crucial to remember that while these symbols look like integers, each represents an entire infinite set of numbers. For example, in  $\mathbb{Z}/12\mathbb{Z}$  (the arithmetic of a clock), the element 2 represents the equivalence class  $[2]$ , which is the set  $\{\dots, -22, -10, 2, 14, 26, \dots\}$ . It is crucial to remember that we are about to define arithmetic not on numbers, but on these infinite sets. The fact that these operations can be defined consistently is a cornerstone of this entire structure and requires rigorous proof.

## 16.3 Operations in $\mathbb{Z}/m\mathbb{Z}$

We can define arithmetic operations on these equivalence classes. For any  $[a], [b] \in \mathbb{Z}/m\mathbb{Z}$ , we define addition and multiplication as follows:

- **Addition:**  $[a] + [b] := [a + b]$
- **Multiplication:**  $[a] \cdot [b] := [a \cdot b]$

A potential issue arises: the result of an operation seems to depend on the specific representatives ( $a$  and  $b$ ) chosen from the classes. The following proposition asserts that this is not the case; the operations are **well-defined**.

**Proposition 16.1** (Compatibility). *Let  $a_1 \equiv a_2 \pmod{m}$  and  $b_1 \equiv b_2 \pmod{m}$ . Then:*

1.  $a_1 + b_1 \equiv a_2 + b_2 \pmod{m}$
2.  $a_1 b_1 \equiv a_2 b_2 \pmod{m}$

This means the result of an operation is independent of the representatives chosen.

*Dimostrazione.* • **Addition:** We are given  $a_2 - a_1 = k_1 m$  and  $b_2 - b_1 = k_2 m$  for some integers  $k_1, k_2$ . Adding these equations yields  $(a_2 - a_1) + (b_2 - b_1) = (k_1 + k_2)m$ . Rearranging the terms, we get  $(a_2 + b_2) - (a_1 + b_1) = (k_1 + k_2)m$ . This shows  $m \mid ((a_2 + b_2) - (a_1 + b_1))$ , so  $[a_1 + b_1] = [a_2 + b_2]$ . Thus, addition is well-defined.

- **Multiplication:** We can write  $a_2 = a_1 + k_1 m$  and  $b_2 = b_1 + k_2 m$ . Multiplying these gives:

$$a_2 b_2 = (a_1 + k_1 m)(b_1 + k_2 m) = a_1 b_1 + a_1 k_2 m + b_1 k_1 m + k_1 k_2 m^2$$

Factoring out  $m$  yields  $a_2 b_2 = a_1 b_1 + m(a_1 k_2 + b_1 k_1 + k_1 k_2 m)$ . This shows that  $a_2 b_2 - a_1 b_1$  is a multiple of  $m$ , so  $[a_1 b_1] = [a_2 b_2]$ . Thus, multiplication is well-defined. □

With these well-defined operations, we can now examine the algebraic structures they create.

## 17 The Algebraic Landscape of $\mathbb{Z}/m\mathbb{Z}$

The operations of addition and multiplication endow the set  $\mathbb{Z}/m\mathbb{Z}$  with a rich algebraic structure. By employing the language of abstract algebra—specifically, the concepts of groups and rings—we can precisely classify its properties and understand its behavior.

### 17.1 The Additive Group $(\mathbb{Z}/m\mathbb{Z}, +)$

**Proposition 17.1.** *The set of integers modulo  $m$  under the operation of addition, denoted  $(\mathbb{Z}/m\mathbb{Z}, +)$ , forms a **commutative group**.*

This structure adheres to the following defining properties:

- **Commutativity:**  $[a] + [b] = [a + b] = [b + a] = [b] + [a]$ .

- **Associativity:**  $([a] + [b]) + [c] = [a] + ([b] + [c])$ .
- **Neutral Element:** The class  $[0]$  is the identity element.
- **Additive Inverse:** For every element  $[a]$ , the opposite element is  $[-a]$ , such that  $[a] + [-a] = [0]$ .

## 17.2 The Commutative Ring $(\mathbb{Z}/m\mathbb{Z}, +, \cdot)$

When we consider both addition and multiplication together,  $(\mathbb{Z}/m\mathbb{Z}, +, \cdot)$  forms a **commutative ring**. This means that  $(\mathbb{Z}/m\mathbb{Z}, +)$  is a commutative group, and the multiplication operation satisfies:

- **Commutativity**
- **Associativity**
- **Multiplicative Neutral Element:** The class  $[1]$  is the identity.
- The Distributive Property holds.

However,  $\mathbb{Z}/m\mathbb{Z}$  is not always a **field**. A key distinction of a field is that every non-zero element must have a multiplicative inverse. For example, consider the element  $[2]$  in  $\mathbb{Z}/4\mathbb{Z}$ . Testing all possible products, we find no inverse:

$$\begin{aligned} [2] \cdot [0] &= [0] \\ [2] \cdot [1] &= [2] \\ [2] \cdot [2] &= [0] \\ [2] \cdot [3] &= [2] \end{aligned}$$

None of these products yield the multiplicative identity  $[1]$ . Therefore,  $[2]$  has no multiplicative inverse in  $\mathbb{Z}/4\mathbb{Z}$ , which motivates a rigorous investigation into which elements possess this crucial property of invertibility.

## 18 Multiplicative Inverses and the Group of Units $(\mathbb{Z}/m\mathbb{Z})^*$

The existence of a multiplicative inverse is a pivotal property in any algebraic system. It is what permits the operation of division and enables the straightforward solution of linear equations, making it a central concept in modular arithmetic.

### 18.1 Defining and Characterizing Invertibility

**Definition 18.1.** An element  $[a']$  is the **multiplicative inverse** of  $[a]$  in  $\mathbb{Z}/m\mathbb{Z}$  if their product is the multiplicative identity:  $[a] \cdot [a'] = [1]$ . The inverse is often denoted  $[a]^{-1}$ .

**Theorem 18.1.** If an inverse exists, it is unique.

*Dimostrazione.* Assume an element  $[a]$  has two inverses,  $[a']$  and  $[a'']$ . By definition,  $[a] \cdot [a'] = [1]$  and  $[a] \cdot [a''] = [1]$ . This implies  $[a] \cdot [a'] = [a] \cdot [a'']$ . The proof can be completed by showing that this requires  $[a'] = [a'']$ .  $\square$

**Theorem 18.2** (Characterization of Invertibility). *An element  $a \in \mathbb{Z}/m\mathbb{Z}$  is invertible if and only if the greatest common divisor of  $a$  and  $m$  is 1:  $\gcd(a, m) = 1$ .*

*Dimostrazione.* An element  $[a]$  is invertible if and only if there exists an integer  $u$  such that  $au \equiv 1 \pmod{m}$ . This congruence is equivalent to the linear Diophantine equation  $au + mv = 1$  (where  $v = -k$ ). By **Bézout's identity**, this equation has an integer solution if and only if  $\gcd(a, m)$  divides 1, which implies  $\gcd(a, m) = 1$ .  $\square$

## 18.2 Finding the Inverse

The proof of the characterization theorem points directly to the **Extended Euclidean Algorithm** as the method for finding the inverse  $u$  by solving  $au + mv = 1$ .

**Example:** Calculate  $5^{-1}$  in  $\mathbb{Z}/7\mathbb{Z}$ .

We solve  $5u + 7v = 1$ .

1. **Euclidean Algorithm:**  $7 = 5 \cdot 1 + 2$ ;  $5 = 2 \cdot 2 + 1$ .

2. **Back-Substitution:**

$$1 = 5 - 2 \cdot 2$$

$$1 = 5 - 2 \cdot (7 - 5 \cdot 1)$$

$$1 = 3 \cdot 5 - 2 \cdot 7$$

The equation  $(3) \cdot 5 + (-2) \cdot 7 = 1$  yields  $u = 3$ . Therefore,  $5^{-1} \equiv 3 \pmod{7}$ .

## 18.3 The Group of Units $(\mathbb{Z}/m\mathbb{Z})^*$

**Definition 18.2.** *The set of all invertible elements (units) in  $\mathbb{Z}/m\mathbb{Z}$  is denoted  $(\mathbb{Z}/m\mathbb{Z})^*$ . Formally:*

$$(\mathbb{Z}/m\mathbb{Z})^* = \{a \in \mathbb{Z}/m\mathbb{Z} : \gcd(a, m) = 1\}$$

**Proposition 18.1.** *The set of units under multiplication,  $((\mathbb{Z}/m\mathbb{Z})^*, \cdot)$ , forms a **commutative group** (the group of units modulo  $m$ ).*

- For  $m = 5$  (prime),  $(\mathbb{Z}/5\mathbb{Z})^* = \{1, 2, 3, 4\}$ .
- For  $m = 12$ ,  $(\mathbb{Z}/12\mathbb{Z})^* = \{1, 5, 7, 11\}$ .

## 19 The Special Case of Prime Moduli: The Field $\mathbb{F}_p$

When the modulus  $m$  is a prime number  $p$ , the algebraic structure  $\mathbb{Z}/p\mathbb{Z}$  simplifies significantly, becoming a **field**, where division is universally possible (except by zero).



## 19.1 Definition and Properties of a Field

**Proposition 19.1.** *For a prime number  $p$ , the commutative ring  $(\mathbb{Z}/p\mathbb{Z}, +, \cdot)$  is a **field**, commonly denoted  $\mathbb{F}_p$ .*

This is because  $\gcd(a, p) = 1$  for all non-zero elements  $a$ , ensuring every element is invertible. The properties of  $\mathbb{F}_p$  include:

- $(\mathbb{F}_p, +)$  is a commutative group.
- $(\mathbb{F}_p \setminus \{0\}, \cdot)$  is a commutative group (the group of units).
- The distributive property holds.

## 19.2 A Key Property of Fields

**Corollary 19.1.** *Let  $p > 2$  be a prime. The equation  $x^2 = 1$  in  $\mathbb{F}_p$  has exactly two distinct solutions:  $x = 1$  and  $x = p - 1$  (i.e.,  $x \equiv -1 \pmod{p}$ ).*

*Dimostrazione.* The equation  $x^2 \equiv 1 \pmod{p}$  is equivalent to  $p \mid (x^2 - 1)$ , or  $p \mid (x - 1)(x + 1)$ . Since  $p$  is prime, by Euclid's Lemma, either  $p \mid (x - 1)$  (implying  $x \equiv 1$ ) or  $p \mid (x + 1)$  (implying  $x \equiv -1 \equiv p - 1$ ). These two solutions are distinct for  $p > 2$ .  $\square$

## 20 Application I: Solving Linear Congruences

The algebraic theory provides a framework for solving linear congruences of the form  $ax \equiv b \pmod{m}$ .

### 20.1 The General Condition for Solutions

**Proposition 20.1.** *The linear congruence  $ax \equiv b \pmod{m}$  has at least one integer solution for  $x$  if and only if  $d = \gcd(a, m)$  divides  $b$ .*

*Dimostrazione.* The congruence is equivalent to the linear Diophantine equation  $ax + mv = b$ . This equation is solvable if and only if  $\gcd(a, m)$  divides  $b$ .  $\square$

### 20.2 Case 1: Unique Solution ( $\gcd(a, m) = 1$ )

When  $\gcd(a, m) = 1$ , the unique solution modulo  $m$  is found by multiplying by the inverse  $a^{-1}$ :

$$x \equiv a^{-1}b \pmod{m}$$

- **Example:** To solve  $245x \equiv 24 \pmod{677}$ , since 677 is prime,  $245^{-1} \equiv 572 \pmod{677}$ .

$$x \equiv 572 \cdot 24 \equiv 13728 \equiv 188 \pmod{677}$$

### 20.3 Case 2: Multiple or No Solutions ( $\gcd(a, m) = d > 1$ )

If  $d = \gcd(a, m)$  does not divide  $b$ , there are no solutions.

**Proposition 20.2.** *If  $d \mid b$ , the congruence  $ax \equiv b \pmod{m}$  has exactly  $d$  distinct solutions modulo  $m$ .*

The method is:

1. **Reduce:** Divide by  $d$ :  $\frac{a}{d}x \equiv \frac{b}{d} \pmod{\frac{m}{d}}$ .
2. **Solve:** Find the unique solution  $x_0$  to the reduced congruence.
3. **Generate Solutions:** The complete set of  $d$  solutions is:

$$x_k = x_0 + k \left( \frac{m}{d} \right) \quad \text{for } k = 0, 1, \dots, d-1$$

- **Example:**  $9x \equiv 6 \pmod{15}$ .
  1.  $d = \gcd(9, 15) = 3$ . Since  $3 \mid 6$ , there are 3 solutions.
  2. **Reduced Congruence:**  $3x \equiv 2 \pmod{5}$ . Solution  $x_0 = 4$ .
  3. **Solutions:**  $m/d = 5$ . Solutions are  $x_0 = 4$ ,  $x_1 = 4 + 5 = 9$ ,  $x_2 = 4 + 10 = 14$ .

## 21 Application II: Shamir's Secret Sharing Scheme

Shamir's Secret Sharing is an elegant application of modular arithmetic and linear algebra for secure information distribution.

## 21.1 The Premise and Analogy

The scheme relies on the principle that  $t$  distinct points uniquely determine a polynomial of degree  $t - 1$ . The secret is encoded as a coefficient (e.g., the y-intercept  $a_0$ ) of a polynomial  $P(x)$ , and each person receives a point  $(x_i, P(x_i))$  on the curve. All calculations are performed in a finite field  $\mathbb{F}_p$ .

## 21.2 A Worked Example

- **Information:** Friend A:  $(23, 401)$ ; Friend B:  $(58, 368)$ .
- **Public Parameter:** Modulus  $p = 941$ .
- **Task:** Calculate the secret slope  $m$  (the secret) of the line passing through the points:  
 $m = (y_2 - y_1) \cdot (x_2 - x_1)^{-1} \pmod{941}$ .

### Calculation

1. **Differences:**  $y_2 - y_1 = -33$ ;  $x_2 - x_1 = 35$ .
2. **Inverse:** Using the Extended Euclidean Algorithm,  $35^{-1} \equiv 242 \pmod{941}$ .
3. **Solve:**  $m \equiv (-33) \cdot 242 \equiv -7986 \pmod{941}$ .
4. **Reduce:**  $-7986 = -9 \times 941 + 483$ .

$$m \equiv 483 \pmod{941}$$

The calculated secret slope is  $m = 483$ .

## 22 An Exposition on the Chinese Remainder Theorem

### 23 Introduction: An Ancient Problem with Modern Relevance

The **Chinese Remainder Theorem (CRT)** is a cornerstone of number theory with a history as rich as its modern applications are diverse. Its origins can be traced to a problem of enumeration found in the 3rd to 5th century Chinese mathematical text, *Sun Tze Suan Ching*. The text poses a now-famous puzzle:

If we count them by threes we have two left over. If we count them by fives we have three left over. If we count them by sevens we have two left over.

This ancient riddle of remainders lays the groundwork for a profound mathematical principle. Centuries later, this principle has found new life as a foundational tool in modern applied mathematics, most notably in the field of cryptography. The theorem provides a powerful framework for solving systems of simultaneous congruences, a task central to the design and implementation of various cryptographic algorithms. This document provides a comprehensive, formal, and academic exploration of the Chinese Remainder Theorem, detailing its formal statement, rigorous proofs, practical solution methodologies, and critical extensions.

### 24 The System of Congruences: Formalizing the Problem

The first step in transforming an abstract puzzle into a solvable mathematical problem is to translate its language into a formal system. This act of formalization is crucial, as it allows us to move beyond a specific case and develop a general, repeatable methodology for all problems of a similar structure. The classic problem described by Sun Tze can be stated as follows:

- If we count them by threes we have two left over.
- If we count them by fives we have three left over.
- If we count them by sevens we have two left over.

In the language of modular arithmetic, this puzzle translates directly into a system of three linear congruences with a single unknown variable,  $x$ :

$$\begin{aligned}x &\equiv 2 \pmod{3} \\x &\equiv 3 \pmod{5} \\x &\equiv 2 \pmod{7}\end{aligned}$$

This formal system is the precise mathematical expression of the original problem. The Chinese Remainder Theorem provides the theoretical framework that guarantees a solution to such a system exists and offers a clear path to finding it.

### 25 The Chinese Remainder Theorem: A Formal Statement

The Chinese Remainder Theorem offers a powerful and elegant guarantee regarding the solution of systems of linear congruences. It asserts that under a specific but critical condition—that the

moduli are pairwise relatively prime—a solution not only exists but is also unique within a well-defined modular range.

**Theorem 25.1** (Chinese Remainder Theorem). *Let  $m_1, m_2, \dots, m_k$  be pairwise relatively prime integers. Let  $a_1, a_2, \dots, a_k$  be arbitrary integers. The system of congruences:*

$$x \equiv a_i \pmod{m_i} \quad \text{for } i = 1 \dots k$$

*has a solution, and the complete set of solutions is congruent modulo the product  $M = m_1 \cdot m_2 \cdot \dots \cdot m_k$ .*

The theorem makes two fundamental promises. The first is **existence**: a solution to the system is guaranteed to exist. The second is **uniqueness**: while there is an infinite set of solutions, they are all congruent to one another modulo  $M$ . This means there is exactly one solution within the range from 0 to  $M - 1$ .

## 26 Solution Methodologies

While the theorem guarantees a solution, several practical methods can be employed to find it. These approaches range from an intuitive, iterative process suitable for smaller systems to a more systematic and scalable algorithm derived directly from the theorem’s constructive proof.

### 26.1 Method 1: Iterative Substitution (“By Hands”)

This intuitive method, suitable for smaller systems, involves solving the first congruence, substituting the resulting expression into the next congruence, and systematically solving for the introduced parameters.

#### Example 1: A 2x2 System

Consider the system:

$$\begin{aligned} x &\equiv 3 \pmod{11} \\ x &\equiv 2 \pmod{7} \end{aligned}$$

1. **Express  $x$ :**  $x = 3 + 11k$ .
2. **Substitute:**  $3 + 11k \equiv 2 \pmod{7}$ .
3. **Solve for  $k$ :**  $4k \equiv -1 \equiv 6 \pmod{7}$ . The inverse of 4 modulo 7 is 2. Multiplying by 2 yields  $k \equiv 2 \cdot 6 \equiv 12 \equiv 5 \pmod{7}$ .
4. **Express  $k$ :**  $k = 5 + 7L$ .
5. **Substitute back:**  $x = 3 + 11(5 + 7L) = 58 + 77L$ .

Thus, the solution is  $x \equiv 58 \pmod{77}$ .

### Example 2: The Sun Tze Problem

$$x \equiv 2 \pmod{3}$$

$$x \equiv 3 \pmod{5}$$

$$x \equiv 2 \pmod{7}$$

1.  $x = 2 + 3k$ . Substitute into second:  $3k \equiv 1 \pmod{5}$ . Inverse of 3 is 2, so  $k \equiv 2 \pmod{5}$ .
2.  $k = 2 + 5m$ . Substitute back into  $x$ :  $x = 2 + 3(2 + 5m) = 8 + 15m$ .
3. Substitute into third:  $8 + 15m \equiv 2 \pmod{7}$ . Since  $15 \equiv 1$  and  $8 \equiv 1 \pmod{7}$ , we get  $1 + m \equiv 2 \pmod{7} \Rightarrow m \equiv 1 \pmod{7}$ .
4.  $m = 1 + 7L$ . Substitute back:  $x = 8 + 15(1 + 7L) = 23 + 105L$ .

The smallest positive solution is  $x = 23$ .

## 26.2 Method 2: The Constructive Algorithm

This algorithm, derived from the theorem's constructive proof, is more robust and scalable for larger systems.

1. **Step 1:** Define the moduli  $m_i$  and calculate their product  $M = m_1 \cdot \dots \cdot m_k$ .
2. **Step 2:** For each  $i$ , calculate  $N_i = M/m_i$ .
3. **Step 3:** Find the modular multiplicative inverse  $v_i$  of  $N_i$  modulo  $m_i$ , such that  $v_i N_i \equiv 1 \pmod{m_i}$ . This is guaranteed by  $\gcd(N_i, m_i) = 1$ .
4. **Step 4 (Solution):** The final solution  $x$  is constructed by the sum:

$$x = \sum_{i=1}^k a_i v_i N_i$$

### Worked Example

Solve the system:

$$x \equiv 2 \pmod{3}$$

$$x \equiv 3 \pmod{7}$$

$$x \equiv 4 \pmod{16}$$

$a_i : 2, 3, 4$ .  $m_i : 3, 7, 16$ .  $M = 3 \cdot 7 \cdot 16 = 336$ .

1. **Calculate  $N_i$ :**  $N_1 = 112$ ,  $N_2 = 48$ ,  $N_3 = 21$ .

2. **Find Inverses  $v_i$ :**

- $112v_1 \equiv 1 \pmod{3} \Rightarrow 1v_1 \equiv 1 \Rightarrow v_1 = 1$ .
- $48v_2 \equiv 1 \pmod{7} \Rightarrow 6v_2 \equiv 1 \Rightarrow v_2 = -1 \equiv 6$ .

$$\bullet \ 21v_3 \equiv 1 \pmod{16} \Rightarrow 5v_3 \equiv 1 \Rightarrow v_3 = -3 \equiv 13.$$

### 3. Construct Solution:

$$\begin{aligned} x &= 2(1)(112) + 3(6)(48) + 4(13)(21) \\ x &= 224 + 864 + 1092 = 2180 \end{aligned}$$

4. **Reduce:**  $x \equiv 2180 \equiv 164 \pmod{336}$ .

The smallest positive solution is  $x = 164$ .

## 27 Formal Proof of the Chinese Remainder Theorem

The proof is divided into two parts: Existence (constructive) and Uniqueness (based on coprimality).

### 27.1 Proof of Existence

We must show that  $x = \sum a_j v_j N_j$  is a solution. We verify that  $x \equiv a_i \pmod{m_i}$  for an arbitrary  $i$ . The argument hinges on the fact that for any  $j \neq i$ , the term  $N_j$  contains  $m_i$  as a factor, leading to:

$$a_j v_j N_j \equiv 0 \pmod{m_i} \quad \text{for } j \neq i$$

Thus, when  $x$  is taken modulo  $m_i$ , all terms vanish except the  $i$ -th term:

$$x \equiv a_i v_i N_i \pmod{m_i}$$

Since  $v_i N_i \equiv 1 \pmod{m_i}$  by construction, the proof concludes:

$$x \equiv a_i(1) \equiv a_i \pmod{m_i}$$

This confirms the existence of a solution.

### 27.2 Proof of Uniqueness

1. **Assume Two Solutions:** Let  $x$  and  $x'$  both solve the system.
2. **Relationship:** For every  $i$ ,  $x \equiv a_i \pmod{m_i}$  and  $x' \equiv a_i \pmod{m_i}$ , implying  $x - x' \equiv 0 \pmod{m_i}$ . Thus,  $m_i$  divides  $(x - x')$ .
3. **Use Co-prime Condition:** Since all  $m_i$  are pairwise co-prime and each divides  $(x - x')$ , their product  $M = m_1 \cdots m_k$  must also divide  $(x - x')$ .
4. **Conclusion:** By definition,  $M \mid (x - x')$  means  $x \equiv x' \pmod{M}$ .

This confirms that the solution is unique modulo  $M$ .

## 28 An Important Extension: The Case of Non-Co-prime Moduli

The powerful guarantees of the CRT rely entirely on the condition that the moduli are pairwise co-prime. When this fails, a solution is not guaranteed.

### Example of No Solution

The system  $x \equiv 0 \pmod{4}$  and  $x \equiv 1 \pmod{6}$  has no solution because  $x$  must be both even (from  $x \equiv 0 \pmod{4}$ ) and odd (from  $x \equiv 1 \pmod{6}$ ), a contradiction.

### Compatibility Condition

For a system of congruences with non-co-prime moduli, a solution exists **if and only if** the following compatibility condition is met for every pair of equations:

$$a_i \equiv a_j \pmod{\gcd(m_i, m_j)} \quad \text{for every pair } i, j$$

If this condition holds, a unique solution exists modulo the **least common multiple** (LCM) of the moduli.

## 29 Conclusion

The Chinese Remainder Theorem is a profound result in number theory that provides a powerful bridge between modular arithmetic with large composite numbers and systems of simpler congruences. Its core function is to transform a complex problem into a manageable set of smaller, independent problems. This “divide and conquer” approach, rooted in an ancient counting puzzle, has become an indispensable tool in modern fields. In cryptography, for example, algorithms like RSA leverage the CRT to dramatically speed up computations involving large moduli. By breaking a single, computationally expensive exponentiation into several smaller ones that can be executed in parallel, the CRT demonstrates its enduring relevance, transforming an elegant piece of ancient mathematics into a cornerstone of modern computational security.



## 30 A Comprehensive Treatise on Euler's Totient Function

### 31 Introduction and Formal Definition

Within the field of number theory, certain functions serve as foundational pillars, providing deep insights into the structure of the integers. Among these, **Euler's totient function**, often denoted by the Greek letter phi ( $\varphi$ ), holds a distinguished place. Its primary role is to quantify the multiplicative structure of integers modulo  $m$ , providing a precise measure of the elements that possess multiplicative inverses. This function, therefore, serves as a direct bridge between the factorization of an integer and the algebraic properties of its corresponding modular system. Formally, for any natural number  $m \geq 1$ , Euler's totient function,  $\varphi(m)$ , can be defined in two equivalent ways:

1. **Algebraic Definition:**  $\varphi(m)$  is the number of elements in the multiplicative group of integers modulo  $m$ , denoted  $(\mathbb{Z}/m\mathbb{Z})^*$ . In other words, it counts the number of invertible elements in the ring  $\mathbb{Z}/m\mathbb{Z}$ .
2. **Combinatorial Definition:**  $\varphi(m)$  is the number of positive integers  $a$  such that  $1 \leq a \leq m$  which are coprime to  $m$ . That is:

$$\varphi(m) = |\{a \in \mathbb{Z} \mid 1 \leq a \leq m, \gcd(a, m) = 1\}|$$

To illustrate this definition, we consider two examples.

- $\varphi(1) = 1$ , since  $\gcd(1, 1) = 1$ .
- $\varphi(14) = 6$ , since the set of coprime integers is  $\{1, 3, 5, 9, 11, 13\}$ .

### 32 Fundamental Properties and Characterization of Primes

Before deriving a general formula for  $\varphi(m)$ , it is instructive to analyze the function's behavior for the special case of prime numbers.

**Theorem 32.1.** *The Totient of a Prime Number: If  $m \geq 2$  is a prime number, then  $\varphi(m) = m - 1$ .*

**Theorem 32.2.** *A Primality Test via the Totient Function: For an integer  $m \geq 2$ ,  $\varphi(m) = m - 1$  if and only if  $m$  is a prime number.*

*Dimostrazione.* ( $\Rightarrow$ ) (**"Only if" direction**): We prove the contrapositive. Assume  $m$  is not prime. By definition,  $m$  must have a non-trivial divisor  $p$  such that  $1 < p < m$ . Since  $p$  divides  $m$ ,  $\gcd(p, m) = p > 1$ . Thus,  $p$  is not coprime to  $m$ . Since  $p$  is an integer between 1 and  $m - 1$  that is excluded from the count, the total count  $\varphi(m)$  must be strictly less than  $m - 1$ . More precisely,  $\varphi(m) \leq m - 2$ . This shows that the equality  $\varphi(m) = m - 1$  can only hold if  $m$  is prime.

( $\Leftarrow$ ) (**"If" direction**): If  $m$  is prime, then  $\varphi(m) = m - 1$ , as established in Theorem ??.

□

### 33 A Formula for Powers of Primes

The formula for calculating the totient of a prime power is the essential building block for constructing the general computational formula for any integer  $n$ .

**Proposition 33.1.** *For any prime  $p$  and any integer  $\alpha \geq 1$ :*

$$\varphi(p^\alpha) = p^\alpha - p^{\alpha-1}$$

*Dimostrazione.* The proof proceeds by counting the complement. An integer  $a$  in the range  $[1, p^\alpha]$  is **not** coprime to  $p^\alpha$  if and only if  $p$  divides  $a$ . We count the number of multiples of  $p$  in the range  $[1, p^\alpha]$ . These multiples are of the form  $k \cdot p$ , where  $k$  is a positive integer. The condition  $1 \leq k \cdot p \leq p^\alpha$  implies  $1 \leq k \leq p^{\alpha-1}$ . Thus, there are exactly  $p^{\alpha-1}$  such multiples of  $p$  in the specified range. Subtracting this count from the total number of integers,  $p^\alpha$ , yields:

$$\varphi(p^\alpha) = (\text{Total Integers}) - (\text{Non-Coprime Integers}) = p^\alpha - p^{\alpha-1}$$

□

- **Example:**  $\varphi(8) = \varphi(2^3) = 2^3 - 2^{3-1} = 8 - 4 = 4$ . (The coprime integers are  $\{1, 3, 5, 7\}$ ).

### 34 The Multiplicative Property of the Phi Function

The central theorem governing the computation of Euler's totient function is its multiplicative property. An arithmetic function  $f$  is said to be multiplicative if  $f(mn) = f(m)f(n)$  whenever  $m$  and  $n$  are coprime.

**Theorem 34.1.** *Multiplicativity of Euler's Totient Function: If  $m$  and  $n$  are coprime integers ( $\gcd(m, n) = 1$ ), then  $\varphi(mn) = \varphi(m)\varphi(n)$ .*

- **Important Caveat:** The property fails if  $m$  and  $n$  are not coprime. For  $m = 4$  and  $n = 6$ ,  $\varphi(4)\varphi(6) = 2 \cdot 2 = 4$ , but  $\varphi(4 \cdot 6) = \varphi(24) = 8$ . Thus,  $8 \neq 4$ .

### 35 Proofs of the Multiplicative Property

#### 35.1 First Proof: A Combinatorial Approach via Inclusion-Exclusion

This proof demonstrates multiplicativity for  $n = p^\alpha q^\beta$  (where  $p, q$  are distinct primes) by counting its complement using the Principle of Inclusion-Exclusion.

1. **Strategy:** Count integers  $k \in [1, n]$  that are a multiple of  $p$  ( $\mathcal{A}_p$ ) OR a multiple of  $q$  ( $\mathcal{A}_q$ ). The size of this set is  $|\mathcal{A}_p \cup \mathcal{A}_q| = |\mathcal{A}_p| + |\mathcal{A}_q| - |\mathcal{A}_p \cap \mathcal{A}_q|$ .
2. **Cardinality Calculation:**
  - $|\mathcal{A}_p| = n/p = p^{\alpha-1}q^\beta$ .
  - $|\mathcal{A}_q| = n/q = p^\alpha q^{\beta-1}$ .
  - $|\mathcal{A}_p \cap \mathcal{A}_q|$  (multiples of  $pq$ )  $= n/(pq) = p^{\alpha-1}q^{\beta-1}$ .

### 3. Derivation:

$$\begin{aligned}\varphi(n) &= n - (|\mathcal{A}_p| + |\mathcal{A}_q| - |\mathcal{A}_p \cap \mathcal{A}_q|) \\ \varphi(n) &= p^\alpha q^\beta - (p^{\alpha-1} q^\beta + p^\alpha q^{\beta-1} - p^{\alpha-1} q^{\beta-1})\end{aligned}$$

### 4. Conclusion (Factoring):

$$\varphi(n) = (p^\alpha - p^{\alpha-1}) \cdot (q^\beta - q^{\beta-1}) = \varphi(p^\alpha) \cdot \varphi(q^\beta)$$

This final expression confirms  $\varphi(n) = \varphi(p^\alpha) \cdot \varphi(q^\beta)$ , completing the combinatorial proof.

## 35.2 Second Proof: An Algebraic Approach via the Remainder Map

This proof uses the algebraic structure of rings and the Chinese Remainder Theorem (CRT).

### 35.2.1 The Remainder Map

We define the remainder map  $r$ :

$$\begin{aligned}r : \mathbb{Z}/(mn)\mathbb{Z} &\rightarrow (\mathbb{Z}/m\mathbb{Z}) \times (\mathbb{Z}/n\mathbb{Z}) \\ r([a]_{mn}) &:= ([a]_m, [a]_n)\end{aligned}$$

This map is well-defined.

### 35.2.2 Bijectivity and the Chinese Remainder Theorem

**Theorem 35.1.** *The remainder map  $r$  is a bijection if and only if  $m$  and  $n$  are coprime.*

- **Surjectivity** (Existence): Guaranteed by the CRT, as every pair of remainders  $([a]_m, [b]_n)$  corresponds to an integer  $x$  (a solution).
- **Injectivity** (Uniqueness): Guaranteed by the CRT, as the solution  $x$  is unique modulo  $mn$ .

### 35.2.3 Mapping Invertible Elements (Units)

An integer  $a$  is coprime to the product  $mn$  if and only if  $a$  is coprime to  $m$  and  $a$  is coprime to  $n$ . This means the bijective map  $r$  restricts to a bijection between the set of units:

$$(\mathbb{Z}/(mn)\mathbb{Z})^* \longleftrightarrow (\mathbb{Z}/m\mathbb{Z})^* \times (\mathbb{Z}/n\mathbb{Z})^*$$

### 35.2.4 Final Derivation

Since there is a one-to-one correspondence (bijection) between the sets of units, their cardinalities must be equal:

$$|(\mathbb{Z}/(mn)\mathbb{Z})^*| = |(\mathbb{Z}/m\mathbb{Z})^*| \times |(\mathbb{Z}/n\mathbb{Z})^*|$$

By the algebraic definition of the  $\varphi$  function, this proves:

$$\varphi(mn) = \varphi(m)\varphi(n)$$

## 36 The General Formula and Conclusion

The findings can be synthesized into a single, powerful method for computing Euler's totient function for any integer. Given  $n$  with its unique prime factorization  $n = p_1^{\alpha_1} \cdot p_2^{\alpha_2} \cdot \dots \cdot p_m^{\alpha_m}$ :

1. **Product of Prime Powers:** Apply the multiplicative property repeatedly:

$$\varphi(n) = \varphi(p_1^{\alpha_1}) \cdot \varphi(p_2^{\alpha_2}) \cdot \dots \cdot \varphi(p_m^{\alpha_m})$$

2. **General Computational Formula (Two Forms):** Substituting the formula from Proposition ?? yields:

$$\varphi(n) = \prod_{i=1}^m (p_i^{\alpha_i} - p_i^{\alpha_i-1})$$

$$\varphi(n) = n \cdot \prod_{i=1}^m \left(1 - \frac{1}{p_i}\right)$$

### Examples of Application

- **Compute  $\varphi(3456)$ :** Factorization  $3456 = 2^7 \cdot 3^3$ .

$$\begin{aligned}\varphi(3456) &= \varphi(2^7) \cdot \varphi(3^3) \\ &= (2^7 - 2^6) \cdot (3^3 - 3^2) \\ &= (128 - 64) \cdot (27 - 9) \\ &= 64 \cdot 18 = 1152\end{aligned}$$

In conclusion, Euler's totient function is a fundamental tool in number theory, providing a direct and computable link between the prime factorization of an integer and the size of its multiplicative group of residues. Its properties, particularly its characterization of primes and its multiplicativity, make it indispensable in the study of modular arithmetic and abstract algebra.